

Notes: Froot, Scharfstein & Stein (JF, 1992)

Herd on the Street: Informational Inefficiencies in a Market with Short-Term Speculation

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1 Big picture

- **Question.** How do short trading horizons change what information speculators choose to learn and trade on?
- **Core contrast.** Standard information-based asset-pricing models imply that informed traders want unique information. Froot, Scharfstein, and Stein show that this result relies on a long-horizon assumption.
- **Main mechanism.** A short-horizon trader profits only if the information he buys becomes incorporated into price before he exits. Therefore, he wants other traders to know and trade on the same information soon after he enters.
- **Central result.** Short horizons turn informational spillovers from negative into positive. This can generate herding on the same signal, multiple research equilibria, and under-diversification of information acquisition.
- **Most striking implication.** Speculators may rationally learn and trade on information unrelated to fundamentals if enough other speculators also trade on it. Chartism can be self-sustaining because it moves short-run prices.
- **Important distinction.** Prices are Bayesian conditional on the order flow; the inefficiency is not irrational pricing. The inefficiency is in the private allocation of research effort before trade.
- **Why this matters.** Short-term speculation can make markets informationally inefficient even when all traders are rational and market makers price competitively.
- **Broader implication.** If firms learn from market prices, short-horizon speculation can distort real decisions by inducing markets to focus on narrow or irrelevant short-run signals.

2 Incremental contribution of the paper

2.1 What the paper adds relative to standard informed-trading models

Most canonical models of asymmetric-information trading, such as Grossman, Hellwig, Verrecchia, and Kyle-type frameworks, emphasize the private value of being one of the few traders who knows a signal. In those models, other informed traders reduce profits because they make price more informative before the trader can exploit the signal. The incremental contribution of this paper is to show that this conclusion is not inherent to rational information acquisition. It depends on the trader's horizon.

- With **long horizons**, the trader earns from the asset's final fundamental payoff. If other traders already know the same signal, the signal is already in price and the trader earns less.
- With **short horizons**, the trader earns from resale to the market before final value is publicly revealed. Other traders using the same signal push that signal into the exit price, which can raise the trader's payoff.
- Therefore, the same rational-expectations environment can produce either contrarian information acquisition or herding depending on the horizon.

Interpretation: The paper's key innovation is to shift the focus from "how much information is in price?" to "what information do traders endogenously choose to produce?" The answer changes once traders care about short-run resale prices rather than final payoffs.

2.2 What the paper adds relative to herding and bubble models

Earlier models can generate nonfundamental price movements through irrational noise traders, positive-feedback traders, or rational bubbles. Froot, Scharfstein, and Stein generate a different form of inefficiency:

- Agents are rational.
- Market makers use Bayes' rule.
- Prices remain conditional expectations given the information embedded in order flow.
- The inefficiency comes from strategic complementarity in information production.

Interpretation: This makes the model conceptually powerful: even without irrational beliefs or mechanical trend chasing, short-horizon incentives can make traders coordinate on a narrow signal or even a non-fundamental signal.

2.3 What the paper contributes to finance and corporate investment theory

The paper also connects speculative horizons to real corporate behavior. If investors focus on information that is useful for short-run price movements, market prices may fail to aggregate the information firms need for long-run investment decisions.

- Existing corporate short-termism models often assume an exogenous information asymmetry between managers and markets.
- This paper provides a mechanism through which that asymmetry can arise endogenously.
- Capital markets may rationally underproduce information about long-term fundamentals because short-horizon speculators do not expect to profit from it.

Takeaway: The incremental contribution is not only a model of “herding on Wall Street.” It is a theory of why the market’s information production technology can be distorted by the horizons of traders, with potential consequences for asset prices and real decisions.

3 Model primitives and measurement

3.1 Risky asset and sources of information

The economy has one risky asset in fixed supply. The asset pays a liquidating dividend:

$$v = a + b. \tag{1}$$

The variables a and b are normally distributed components of fundamental value. In the symmetric benchmark, a and b are equally informative about v .

- A speculator can learn either a or b , but not both.
- The total number of speculators is n .
- n_a speculators observe a and n_b observe b .
- In the efficient symmetric benchmark, $n_a = n_b$ when a and b have equal variances.

Interpretation: Learning is the real choice variable. The paper treats information production as endogenous and asks whether decentralized speculators choose the socially informative allocation.

3.2 Trader types

There are three classes of traders.

- **Speculators** acquire one signal and submit market orders.
- **Liquidity traders** submit noisy inelastic orders unrelated to value.
- **Market makers** are risk neutral, competitive, and set prices equal to conditional expectations of v given order flow.

Interpretation: Liquidity traders prevent prices from fully revealing private information immediately. Without liquidity noise, the return to becoming informed would be eliminated because prices would perfectly reveal the signal.

3.3 Timing and horizon parameter

At date 0, speculators choose which signal to learn. There are then three trading dates.

- At dates 1 and 2, speculators' orders are executed. Half are executed early and half late.
- At date 3, speculators unwind their positions.
- With probability α , the dividend v is publicly announced at date 3.
- With probability $1 - \alpha$, v is not revealed until date 4, after speculators have exited.

The parameter α measures speculative horizon length.

- α close to 1: long horizons; traders are likely to exit after fundamentals are revealed.
- α close to 0: short horizons; traders are likely to exit before fundamentals are revealed.

Interpretation: The central comparative static is a change in α . It determines whether a trader cares mainly about final value or about the interim price at which he can resell.

3.4 Order flow and market-maker pricing

Order flow at date $t = 1, 2$ combines informed orders and liquidity noise:

$$F_t = \frac{n_a}{2}q_a + \frac{n_b}{2}q_b + \varepsilon_t, \quad (2)$$

where q_a and q_b are equilibrium demands of speculators informed about a and b .

Market makers set prices as conditional expectations. With normality, the date-1 price is linear in order flow:

$$p_1 = A_1 F_1, \quad A_1 = \frac{\text{Cov}(v, F_1)}{\text{Var}(F_1)}. \quad (3)$$

At date 2, the price uses the average of the two order flows:

$$p_2 = A_2 \frac{F_1 + F_2}{2}. \quad (4)$$

Because two periods of order flow are more informative than one, $A_2 > A_1$.

Interpretation: The fact that $A_2 > A_1$ is what lets early short-horizon traders profit from later traders using the same information. Their signal becomes more embedded in price between entry and exit.

3.5 Informational efficiency benchmark

The paper defines informational efficiency as correct allocation of research effort, not merely rational pricing. In the symmetric benchmark with equally informative a and b , the efficient allocation is:

$$n_a = n_b. \quad (5)$$

One benchmark objective is to minimize the average variance of prices around true value:

$$\frac{1}{2}E[(v - p_1)^2] + \frac{1}{2}E[(v - p_2)^2]. \quad (6)$$

Interpretation: The market can be Bayesian conditional on the information traders produce and still be informationally inefficient if traders choose the wrong information to produce.

3.6 Chartist extension

The paper also introduces a variable c that is independent of fundamentals. Traders who study v are called fundamentalists; traders who study c are called chartists.

- Fundamentalists learn true payoff-relevant information.
- Chartists learn pure noise.
- Chartists can still profit if enough other chartists also trade and move prices before exit.

Interpretation: This extension is the sharpest result in the paper. It shows that rational traders can choose to study nonfundamental variables because those variables help predict short-run price movements created by other rational traders.

4 Theoretical specifications and propositions

4.1 Speculator payoff from an a signal

A speculator observing a chooses demand q_a^i . If v is revealed at date 3, profits depend on final value. If not, profits depend on the interim price change from p_1 to p_2 . The conditional payoff can be represented as:

$$U_a^i = q_a^i E \left[\alpha \left(v - \frac{p_1 + p_2}{2} \right) + (1 - \alpha) \frac{p_2 - p_1}{2} \mid a \right]. \quad (7)$$

Interpretation: The first term is the long-horizon fundamental payoff. The second term is the short-horizon resale payoff. The model's logic comes from how the relative weight on these terms changes with α .

4.2 Strategic substitutes under long horizons

When α is high, a trader's payoff comes mainly from buying below fundamental value and holding until value is revealed. In this case, other traders using the same signal reduce the trader's profit because their orders already push that signal into the entry price.

$$\frac{\partial U_a^i}{\partial q_a} < 0 \quad \text{when } \alpha \text{ is high.} \quad (8)$$

Interpretation: Information production is contrarian. A trader wants to study what others are not studying because unexploited information generates larger profits.

4.3 Strategic complements under short horizons

When α is low, a trader expects to exit before value is revealed. The trader profits if his information becomes more embedded in price before exit. Other similarly informed traders help make this happen.

$$\frac{\partial U_a^i}{\partial q_a} > 0 \quad \text{when } \alpha \text{ is low.} \quad (9)$$

Interpretation: Information production becomes a coordination game. A signal is valuable partly because others will also trade on it.

4.4 Equilibrium demands and market depth

In a symmetric trading equilibrium, the demand of an a -speculator takes the form:

$$q_a = \delta_a a, \quad (10)$$

where the aggressiveness coefficient δ_a depends on the number of a -speculators and on market-maker pricing coefficients A_1 and A_2 .

The corresponding system also pins down A_1 and A_2 from market-maker Bayesian updating:

$$A_1 = \frac{\text{Cov}(v, F_1)}{\text{Var}(F_1)}, \quad A_2 = \frac{\text{Cov}\left(v, \frac{F_1 + F_2}{2}\right)}{\text{Var}\left(\frac{F_1 + F_2}{2}\right)}. \quad (11)$$

Interpretation: There is a fixed-point structure: speculators' trades determine order-flow informativeness; market makers' price impact determines speculative profits; those profits determine information acquisition.

4.5 Research-stage equilibrium

At date 0, speculators choose whether to learn a or b . An interior research equilibrium must satisfy:

$$\text{EU}_a = \text{EU}_b, \quad (12)$$

and no individual speculator can gain by switching signals. Operationally, at the symmetric point, expected utility must decline with additional same-signal speculators.

Interpretation: Equal expected utility is not enough. Stability requires that the symmetric point be locally self-enforcing. This is exactly where short horizons break the efficient allocation.

4.6 Proposition 1: long horizons support informational efficiency

Proposition 1. If speculators have sufficiently long horizons, the informationally efficient outcome $n_a = n_b$ is an equilibrium of the research game.

Interpretation: With long horizons, each trader dislikes crowding in his signal. If too many traders study a , some want to switch to b . This decentralized force pushes research effort toward information diversity.

4.7 Proposition 2: short horizons produce herding

Proposition 2. If speculators have sufficiently short horizons, the informationally efficient outcome is not an equilibrium of the research game; any research equilibrium involves herding.

Interpretation: With short horizons, each trader values being near others because same-signal trades move the exit price. The decentralized equilibrium therefore overproduces some information and underproduces other information.

4.8 Proposition 3: trading on nonfundamental noise

Let c be independent of fundamentals and let n_c chartists study c , while n_v fundamentalists study v . The paper shows:

Proposition 3. If $n_c > n_v$ and α is sufficiently small, there exists an equilibrium in which both chartists and fundamentalists trade actively and earn positive expected profits. There is also always an equilibrium in which chartists do not trade.

Interpretation: Chartist trading is self-fulfilling. If many chartists trade, the price moves with c in the short run. Individual chartists can then profit from predicting price movements even though c has no fundamental content.

5 Identification and theoretical strategy

5.1 Nature of the strategy

This is a theoretical paper, so identification does not come from empirical variation. The strategy is to isolate the logical role of trading horizons in a controlled rational-expectations model.

- Hold preferences, market-maker rationality, and liquidity noise fixed.
- Vary the horizon parameter α .
- Compare equilibrium information acquisition under long and short horizons.
- Use the symmetric case to define a clean informational-efficiency benchmark.
- Use the chartist extension to test how far the mechanism can go when information is nonfundamental.

Interpretation: The paper's credibility comes from internal model discipline. The same environment produces efficient information diversity when horizons are long and inefficient herding when horizons are short.

5.2 Why the horizon parameter identifies the mechanism

The parameter α changes the probability that value is revealed before speculators exit.

- High α makes final fundamentals important for payoff.
- Low α makes interim price movements important for payoff.
- Because $A_2 > A_1$, same-signal trading increases the short-run exit price sensitivity to the signal.

Interpretation: The sign change in spillovers is the identified mechanism. The model does not need irrationality, borrowing limits, or institutional frictions beyond the short horizon.

5.3 Why liquidity noise matters

Liquidity noise prevents immediate full revelation of information. If market makers could perfectly infer signals from order flow, informed trading profits would vanish. Noise makes prices partially revealing.

Interpretation: Noise is not the source of irrationality. It is a necessary ingredient for rational information acquisition, as in Grossman-Stiglitz and Kyle models.

5.4 Why staggered execution matters

The assumption that some orders execute at date 1 and others at date 2 means information diffuses gradually into price. Early traders can enter before the full price impact of their signal and exit after more same-signal trades occur.

Interpretation: This creates the “beat the gun” incentive. The trader does not merely want to be correct; he wants to be early relative to other traders who will soon move the price.

5.5 Main threats and model limitations

- The model treats horizons as exogenous rather than deriving them from agency frictions or financing constraints.
- Speculators cannot condition their demands on current price in the most realistic way.
- The baseline model restricts entry of new long-horizon informed traders at date 3.
- The welfare benchmark is informational rather than a full production-based welfare criterion.
- The chartist result is theoretical; the paper does not empirically test chartism.

Interpretation: These limitations do not weaken the central logical contribution: once the payoff horizon is short, rational traders may prefer crowded and even nonfundamental signals.

6 Figures and detailed interpretation

6.1 Figure 1: Equilibrium when traders have relatively long horizons

Read:

- The horizontal axis is the number of a -speculators, n_a , holding total speculators fixed at 20.
- The vertical axis plots expected utilities of a - and b -speculators.
- The parameter is $\alpha = 0.25$, so horizons are relatively long.
- The utility curves cross at $n_a = 10$, the symmetric allocation.
- Around the crossing, expected utility falls when more traders study the same signal.

Interpretation: This figure visualizes negative information spillovers. If too many traders study a , an additional a -speculator has lower utility than a b -speculator. The private incentive to switch restores the efficient split of research effort.

Economic message: Long-horizon markets produce contrarian research incentives. Traders seek information that is not yet crowded, so decentralized information acquisition diversifies across fundamentals.

6.2 Figure 2: Equilibrium when traders have relatively short horizons

Read:

- The same utility curves are plotted with a shorter horizon, $\alpha = 0.03$.
- The symmetric point still equalizes utilities, but it is no longer locally stable.
- There are two off-center stable equilibria labeled A and B .
- These equilibria involve herding: most traders study one signal and few study the other.

Interpretation: Short horizons transform the research game into a coordination game. Near the symmetric point, a trader benefits from moving toward the signal that more traders are studying because that signal is more likely to move the near-term exit price.

Economic message: The market remains rational in pricing, but research effort is inefficient. Prices overweight the signal that speculators coordinate on and underweight the neglected signal.

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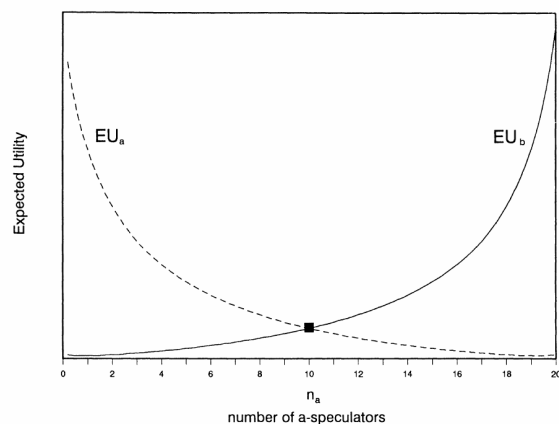


Figure 1. Equilibrium when traders have relatively long horizons. EU_a and EU_b are the expected utilities of a - and b -speculators, respectively. These expected utilities are graphed as a function of the number of a -speculators, n_a , holding the total number of speculators fixed. In the construction of this figure the number of speculators is 20; the variances of a , b , and ε are unity; and α , the probability that the liquidating dividend is announced at date 3, is 0.25.

(a) Long horizons: symmetric information allocation is stable.

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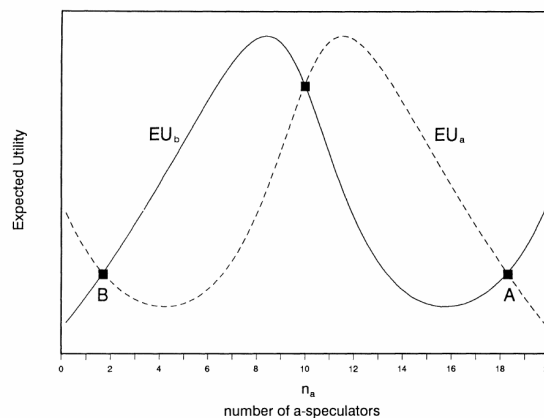


Figure 2. Equilibrium when traders have relatively short horizons. EU_a and EU_b are the expected utilities of a - and b -speculators, respectively. These expected utilities are graphed as a function of the number of a -speculators, n_a , holding the total number of speculators fixed. In the construction of this figure the total number of speculators is 20; the variances of a , b , and ε are unity; and α , the probability that the liquidating dividend is announced at date 3, is 0.03.

(b) Short horizons: herding equilibria appear.

Figures 1 and 2 from the paper: horizon length changes the stability of the efficient information allocation.

6.3 Figure 3: Equilibrium when fundamentalists and chartists have relatively long horizons

Read:

- Fundamentalists learn value-relevant information v .

- Chartists learn pure noise c .
- The horizontal axis is the number of chartists n_c .
- With relatively long horizons, EU_v dominates EU_c for the relevant range.
- Chartists may trade in the trading subgame, but they do not want to choose chartist information at the research stage.

Interpretation: Long horizons punish nonfundamental information. Even if chartist trades move prices temporarily, traders expect final value to matter enough that nonfundamental signals are unattractive.

Economic message: When fundamentals are revealed before exit with sufficient probability, rational speculation disciplines attention toward fundamentals.

6.4 Figure 4: Equilibrium when fundamentalists and chartists have relatively short horizons

Read:

- The same fundamentalist versus chartist environment is shown with a very small α .
- The chartist utility curve intersects the fundamentalist utility curve at point C .
- Point C is a self-sustaining equilibrium with many chartists.
- The chartist signal is not fundamentally informative, but it predicts short-term price movements generated by chartist trading itself.

Interpretation: This is the most striking figure in the paper. The nonfundamental signal becomes valuable because many traders coordinate on it. Traders do not have to believe that chartism predicts cash flows; they only need to believe it predicts interim prices.

Economic message: Short horizons can make nonfundamental research privately profitable and socially wasteful. This helps rationalize market episodes in which attention shifts toward technical, conventional, or fashionable indicators.

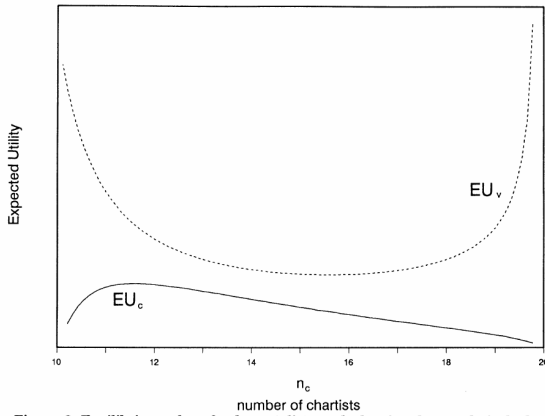


Figure 3. Equilibrium when fundamentalists and chartists have relatively long horizons. EU_v and EU_c are the expected utilities of fundamentalist speculators (those who observe the asset's true value, v) and chartist speculators (those who observe pure noise, c), respectively. These expected utilities are graphed as a function of the number of chartist speculators, n_c , holding the total number of speculators fixed. In the construction of this figure the total number of speculators is 20; the variances of v , c , and ε are unity; and α , the probability that the liquidating dividend is announced at date 9, is 0.01.

(a) Long horizons: fundamentalist research dominates chartism.

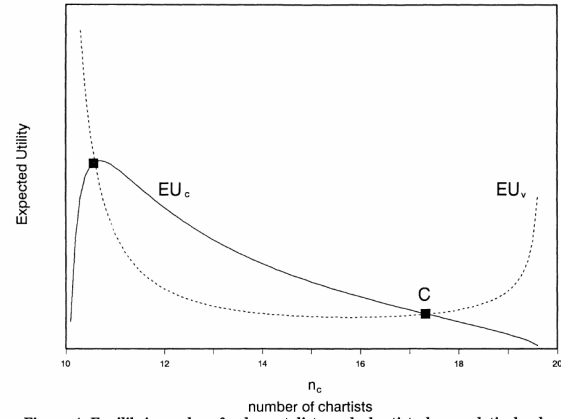


Figure 4. Equilibrium when fundamentalists and chartists have relatively short horizons. EU_v and EU_c are the expected utilities of fundamentalist speculators (those who observe the asset's true value, v) and chartist speculators (those who observe pure noise, c), respectively. These expected utilities are graphed as a function of the number of chartist speculators, n_c , holding the total number of speculators fixed. In the construction of this figure the total number of speculators is 20; the variances of v , c , and ε are unity; and α , the probability that the liquidating dividend is announced at date 3, is 0.001.

(b) Short horizons: chartism can be self-sustaining.

Figures 3 and 4 from the paper: short horizons can support trading on nonfundamental noise.

7 Interpretive synthesis

7.1 The beauty-contest logic in rational expectations form

The Keynesian beauty-contest idea is that traders care about what other traders will think in the near future. The paper formalizes this using a rational-expectations market. A trader who exits before fundamentals are revealed cares about the information that will be impounded into the resale price, not only the information that predicts long-run value.

Takeaway: Beauty-contest behavior need not be irrational. It can arise naturally when the payoff horizon is shorter than the fundamental-resolution horizon.

7.2 Why herding is not necessarily observational irrationality

The model's herding differs from naive imitation. Speculators are not blindly following each other. They rationally choose signals that other traders are likely to use because those signals determine near-term price movements.

Takeaway: Herding here is a rational coordination failure in information acquisition.

7.3 Connection to market efficiency

The paper redefines market efficiency to include the production of information. Prices can be conditionally efficient given the information in order flow and still be inefficient because the set of information that enters order flow is too narrow or irrelevant.

Takeaway: Rational pricing is not sufficient for informational efficiency when information acquisition is endogenous.

7.4 Connection to corporate short-termism

If stock prices guide managerial investment, then the market's research choices matter for real allocation. Short-term traders may focus on signals such as near-term earnings announcements and ignore information about long-term investments, making stock prices less useful for disciplining long-run corporate decisions.

Takeaway: The model provides a capital-market source of real short-termism: not just managers being myopic, but markets failing to produce long-term information.

8 Short summary

- The paper asks how short trading horizons affect information acquisition and price efficiency.
- In standard long-horizon models, informed traders want unique information because other informed traders compete away profits.
- In short-horizon markets, traders profit if their information is incorporated into the price before they exit.
- Therefore, other traders using the same information can raise expected profits.
- This creates positive spillovers and strategic complementarities in information acquisition.
- With long horizons, the efficient symmetric allocation of research effort is stable.
- With short horizons, the efficient allocation can become unstable and herding equilibria appear.
- In the extension, traders may rationally study a pure noise signal if many others also trade on it.
- Prices are Bayesian conditional on order flow, but the information entering the order flow can be socially inefficient.
- The model provides a rational foundation for market attention cycles, chartism, and Keynesian beauty-contest trading.
- The model also suggests a channel linking short-horizon speculation to real corporate short-termism.

9 Conclusion

9.1 Core conclusion

Froot, Scharfstein, and Stein show that short speculative horizons can make rational traders herd on the same information. The mechanism is that a short-horizon trader wants his signal to be impounded into price before he exits. This makes other similarly informed traders complements rather than competitors.

9.2 Incremental lesson

The paper's incremental contribution is to show that market inefficiency can arise before the pricing stage. Market makers price rationally, and speculators optimize rationally. Yet the market may underproduce fundamental information or overproduce nonfundamental information because decentralized research incentives are distorted by short horizons.

9.3 What the paper does not claim

The paper does not claim that all short-horizon trading is inefficient, that all chartism is irrational, or that prices ignore information once it is produced. It also does not estimate the magnitude of the effect in data. Its contribution is a theoretical mechanism showing why short horizons can distort information production.

9.4 Practical implications

- Markets with many short-horizon traders may focus excessively on signals that predict near-term price movements.
- Public information releases can shape private research incentives by changing the horizon over which information remains private.
- Financial institutions, compensation contracts, and delegated asset management structures that shorten trading horizons may affect the informational quality of prices.
- Firms using market prices as signals may be misled when prices reflect attention to narrow or nonfundamental variables.

9.5 Final takeaway

The enduring insight is that the horizon of speculation determines the private value of information. Long-horizon traders want to know what others do not know. Short-horizon traders want to know what others will soon know. That simple change can turn information acquisition into a coordination problem and can make rational markets informationally inefficient.